For Examiner's Use

3 **(a)** An ideal gas is assumed to consist of atoms or molecules that behaves as hard, identical spheres that are in continuous motion and undergo elastic collisions.

State two further assumptions of the kinetic theory of gases.

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.....

..... [2]

(b) Helium-4 ($^{4}_{2}He$) may be assumed to be an ideal gas.

(i) Show that the mass of one atom of helium-4 is 6.6×10^{-27} kg.

[1]

(ii) Determine the internal energy of 1.2 mole of helium-4 gas at a temperature of 27 °C.

internal energy = ... J [1]

For Examiner's Use

(iii) Determine the root-mean-square (r.m.s.) speed of a helium-4 atom at a temperature of 27 °C.

root-mean-square speed = ……. m s^{-1} [2]

(iv) The above r.m.s. speed for helium is less than the escape speed for particles on the Earth's surface and yet there are still some helium atoms that escape the Earth's atmosphere. Explain.

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……..

…… [1]